
Applying Federated Learning to Small-Molecule Datasets

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Abstract

Machine learning has become a critical tool in early-stage drug discovery, where predictive models can substantially reduce the cost and time required to identify promising lead compounds. However, the effectiveness of modern models such as deep neural networks depends on access to large, well-sampled, and well-labeled biochemical datasets. In practice, such data are scarce, expensive to generate, and fragmented across pharmaceutical companies that are unwilling or unable to share proprietary experimental results. Recent federated learning techniques enable multiple parties to learn shared models without data sharing, making them attractive as means for collaboration between drug developers. In particular, vertical federated learning and federated transfer learning methods are most appropriate for the kinds of data held by drug developers which are poorly aligned in feature and sample space. This work tests the FedBCD and FedCVT algorithms on a small molecule classification task, benchmarking against single-party and horizontal federated learning. It finds that the tested vertical and transfer learning methods work just as well if not better than the omniscient, one-party model, leaving open questions for where and how to apply these methods next in pharmaceutical development. Code is available at <https://github.com/vigilius405/pharma-transfer-learning>.

1 Introduction and Motivation

1.1 Machine Learning in Drug Discovery

Drug discovery is a notoriously expensive and time-consuming process, often spanning over a decade and costing over a billion dollars per approved compound [1]. The earliest phase of this pipeline, lead discovery, involves screening vast chemical spaces to identify molecules with desirable biological and pharmacokinetic properties. Machine learning models have shown promise in accelerating this stage by predicting properties such as binding affinity, biocompatibility, and toxicity from molecular structure alone. Such predictions can help narrow the search space significantly, greatly saving time and resources.

Despite this promise, practical deployment of machine learning in drug discovery faces significant limitations. High-performing models typically require training data on the order of millions of compounds, yet experimental assays are expensive and slow. Decades of such research has been compiled into a number of publicly available databases, such as PubChem, UniProt, and GenBank, for different classes of molecules. Yet compared to other modalities machine learning is used for such as text and image analysis and generation, the millions of compounds in these datasets (only hundreds of thousands of which are well-characterized) are pitifully few [1]. Large pharma companies hold

proprietary data on millions more compounds tested for previous projects, data which could help train improved models when used together.

1.2 Data Silos in Pharmaceutical Development

The resulting data silo dilemma poses the question of whether multiple institutions may collaboratively train better models without directly sharing the experimental data they expended significant resources to accumulate. From a business perspective, collaboration at the lead discovery stage may be acceptable, as this phase is relatively inexpensive compared to later-stage clinical trials and does not immediately reveal commercially viable drug candidates [1,2]. From a technical perspective, however, enabling such collaboration is more challenging.

In many realistic settings, data across institutions are not simply partitioned by samples. One company may have toxicity labels for a subset of compounds, while another may have binding affinity measurements for a partially overlapping set. Some compounds may appear in multiple silos with different experimental readouts, while others appear in only one. This structure violates the assumptions of standard horizontal federated learning and motivates the use of more general distributed learning methods.

1.3 Vertical Federated Learning and Federated Transfer Learning

Vertical federated learning (VFL) addresses scenarios in which multiple parties share a common sample space but hold different feature sets, with labels available at only one or a few parties. In VFL, model training proceeds by exchanging intermediate representations or encrypted statistics rather than raw features or labels. Recent work has included efficient optimization schemes to this problem class, such as block coordinate descent-based methods which allow parties to jointly optimize a global objective while preserving data locality [3].

Federated transfer learning (FTL) further generalizes this setting by allowing both feature spaces and sample spaces to differ across parties, with only limited overlap. By incorporating ideas from transfer learning and semi-supervised learning, FTL methods aim to leverage unlabeled or non-overlapping data to improve performance on a target task. Large-scale industrial efforts, such as the MELLODDY project, suggest that these approaches may be viable in pharmaceutical applications. This European Union-funded initiative brought together ten of the world’s largest pharma companies to learn from over 2.6 billion privately generated data points on small molecules [4]. Public evidence of its methods, internal uses, strengths, and weaknesses remain scarce, however. Greater public resources on the use of VFL and FTL for life sciences datasets are therefore desired.

1.4 Aims

The primary goal of this project is to empirically evaluate whether vertical federated learning and federated transfer learning can provide meaningful benefits on small-molecule prediction tasks under realistic data constraints. Specifically, I aimed to:

1. Identify suitable biochemical datasets that reflect the small-sample, heterogeneous nature of real-world pharmaceutical data.
2. Establish strong baselines using single-device and horizontal learning.
3. Evaluate vertical federated learning on shared samples while preserving data silos.
4. Explore federated transfer learning methods that leverage both shared and non-shared samples across parties.

2 Methods

2.1 Datasets

I base these experiments on datasets from the MoleculeNet benchmark, which provides a collection of small to medium-sized molecular property prediction tasks commonly used in scientific machine learning. In contrast to large proprietary datasets, MoleculeNet datasets typically contain between

one and several thousand labeled compounds, making them representative of the low-data regime faced by many research groups [5].

I focus on the physiological classification tasks in MoleculeNet. These are Simplified Molecular Input Line Entry System (SMILES)-based classification tasks related to toxicity, side effects, and drug approval, including BBBP (blood-brain barrier penetration), ClinTox (clinical trial toxicity and FDA approval), SIDER (side effect prediction), Tox21, and ToxCast (both for preclinical toxicology screening). These datasets differ in the label space, enabling evaluation of federated methods across a range of supervision settings [5]. Upon starting this project, it was assumed that there would be some amount of overlap between the sample distributions in these datasets as well, which was explored in accordance with the first project aim. The number of samples found in between one and five of the datasets was characterized, and the Jensen-Shannon distance between the length distributions of SMILES sequences in each of the datasets was used to inform task selection. The goal was for a more challenging task, so the least aligned datasets were used for further study.

2.2 Molecular Representation

A key challenge in applying sequence models to small molecules is converting molecular structure into a suitable input representation. While graph neural networks explicitly model molecular graphs, a recent body of work has demonstrated strong performance from sequence models such as Large Language Models (LLM) trained on SMILES strings [6]. SMILES encodes molecular graphs as linear sequences, enabling the use of such architectures and pretrained embeddings.

In this project, SMILES strings are first canonicalized and aligned to reduce representational variability. I then use the Universal Molecular Encoder (UME, small version), a pretrained model from Prescient Design designed to produce fixed-length embeddings that capture chemical and biological properties of molecules [7]. These embeddings (dimension 384) serve as high-level molecular features that can be combined with auxiliary experimental features where from the respective datasets.

Given molecular embeddings and task-specific features, I employ a simple multilayer perceptron (MLP) as the prediction head. This choice reflects a desire to isolate the effects of distributed learning from those of highly complex architectures. The MLP is trained to predict target labels such as toxicity or FDA approval, depending on the dataset.

2.3 Vertical Learning with FedBCD

For vertical federated learning on shared samples, I adopt the FedBCD (Federated Block Coordinate Descent) algorithm. In this framework, each party maintains local model parameters corresponding to its own feature subset. During training, parties exchange intermediate activations rather than raw data. Specifically, these activations are feature-wise linear transformations of the data with the model weights [8].

Formally, the global objective is defined as an average loss over samples, with regularization applied to each party’s parameters. At each communication round, parties perform multiple local gradient updates using possibly stale intermediate representations from other nodes. This strategy reduces communication overhead while preserving convergence guarantees under standard assumptions [8]. Note that the sequential version of FedBCD was used; FedBCD and FedCVT were implemented as pseudo-distributed (algorithmically equivalent, but performed in a single process).

2.4 Federated Transfer Learning with FedCVT

To handle non-overlapping samples and exploit unlabeled data, I explore federated transfer learning via Cross-View Training (FedCVT). FedCVT extends vertical federated learning by introducing additional prediction and regularization tasks between views held by different parties. Each party learns representations of its own aligned and unaligned features, using losses between party and alignment type to hold a more unified representation together. These representation models are then used to predict missing labels with party-individual and shared classifiers; if there is agreement, these samples are kept and contribute to the classifiers’ ongoing training. Multiple cycles of this process produce confident labels for unaligned samples [9]. In this case, the representation and classifier networks were the same MLPs used for the other methods. The attention module implemented was

as described in the paper, a sharpened scaled dot product attention, and is used to estimate new representations from learned representations.

In sum, the FedCVT approach allows information from unlabeled or partially overlapping samples to influence the learned representations, potentially improving generalization in low-label regimes. In these experiments, FedCVT is applied to settings where only a subset of compounds have labels and feature overlap across parties is limited. Together, these methods enable a systematic comparison of centralized, horizontal (All-Reduce), vertical, and federated transfer learning paradigms on realistic small-molecule datasets.

3 Results

3.1 Data Exploration

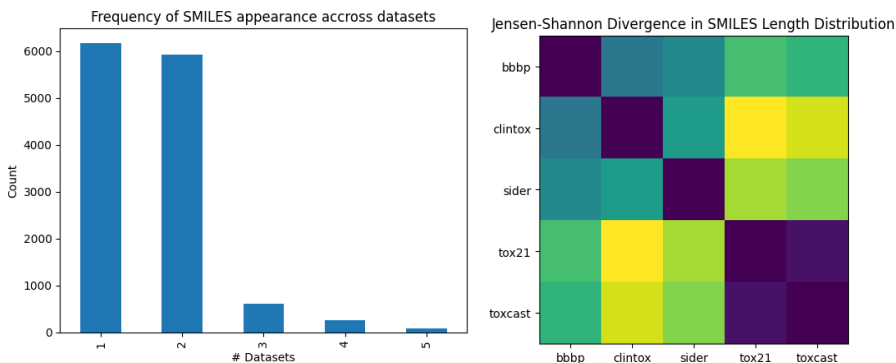


Figure 1: **Left)** Frequency of SMILES sample appearances across the five MoleculeNet datasets. **Right)** Jensen-Shannon distance in SMILES length distribution across datasets (lighter values are more divergent).

As displayed in Figure 1, it was found that the overlap between small molecules in the five datasets was mostly between pairs of datasets, with few molecules being found in more than two of the datasets. The similarity between distributions (or at least a proxy for the distribution, which is SMILES sequence length) is shown on the right panel. Tox21 and Toxcast datasets are the most similar, while Clintox and Tox21 are the most different. To yield a more difficult FTL task, this latter pair was chosen.

Only 485 samples were shared between Clintox and Tox21, leaving 993 unaligned Clintox samples and 7346 unaligned Tox21 samples. These unaligned samples represent the vast majority of data, which could be important to improved learning. The model learning objective (the label) was thus the binary FDA approval category from Clintox, with the other non-sequence predictive feature from Clintox being whether the molecule was known to have toxic effect in humans. The Tox21 dataset consists of 12 categories of preclinical toxicity screening in addition to the molecular SMILES sequence, however many of the compounds are missing (NaN) readouts for these experiments. This fact demonstrates even more how this benchmark is a realistic parallel to the sparse data environment of pharmaceutical development.

3.2 Model Performance

Table 1: Classification Test Accuracy

Model	One Party	All-Reduce	FedBCD	FedBCD Local	FedCVT
Validation Accuracy (Mean)	0.955	0.988	0.974	0.960	0.967
Validation Accuracy (σ)	0.018	0.014	0.018	0.020	0.017
Significance vs. One Party		****	**	n.s.	*

All models were trained ten times, yielding the means and standard deviations of validation accuracy in Table 1. FedBCD local is the version of FedBCD where each party takes a number of local steps (in this case 5) with stale activations from the other parties between synchronizations. Significance was computed using the p-values of hypothesis tests from the means, standard deviations, and sample sizes of the experiments. All-Reduce, FedBCD, and FedCVT showed significantly improved task performance over the one party method. FedCVT did not show significant improvement over FedBCD.

4 Discussion

Other than the local FedBCD algorithm, all the others actually improved over the one party performance. This was surprising; the one party algorithm is essentially omniscient, while the others all rely on limited view of the data. This is likely due to parameter count being more impactful than I had initially expected, as distributed methods each contain individual full models for a smaller shard of the data. This is particularly the case in FedCVT, which utilizes multiple representation, attention, and classifier networks. This ranking would thus potentially change if controlled for parameter count, for example by reducing hidden dimensions in split models.

It was also somewhat surprising that the FedCVT algorithm did not show improved performance over FedBCD, despite making greater use of the available data. It is likely the case that this objective was essentially too easy in that both hit somewhat of an upper limit. Perhaps with a more difficult learning objective, FedCVT would show improved performance. Note also that the all-reduce method does not make much sense in this learning context, since the structure of data silos in drug discovery is not usually compatible with horizontal learning. However this was still tested to provide a distributed benchmark alongside the one party benchmark.

Future directions would see these methods being applied to additional molecular classification tasks and, more importantly, more difficult regression tasks. Bringing in other types of molecular sequences or experimental data would also improve the transferability of these conclusions. Additional VFL and FTL methods would also have to be implemented to gain a broader sense of the method space and more data on what works best for these data and kinds of questions. What is certain is that this work has shown VFL and FTL as viable methods for distributed, private learning from proprietary molecular data.

References

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A Technical Appendix

Table 2: Hyperparameters

Model	One Party	All-Reduce	FedBCD	FedBCD Local	FedCVT
Epochs	100	100	100	100	100
Learning Rate	0.001	0.001	0.001	0.001	0.001
Batch Size	32	32	32	32	32
Input Size	397	397	385, 396	385, 396	385, 396
Hidden Size	256	256	256	256	256
Output Size	1	1	1	1	1
World Size	1	2	2	2	2
Number Local Updates q			1	5	
Pseudo-label Threshold t					0.5
Sharpening Temperature T					0.5
Lambdas					0.1 for all